REQUIREMENTS & INITIAL DESIGN

1. Pick any one app you use daily (Instagram, Zomato, Flipkart, etc.) and imagine you arehired to design its college campus WiFi network — write a requirements document listing at least 8 must-have features or needs from the perspective of students using that app on campus.

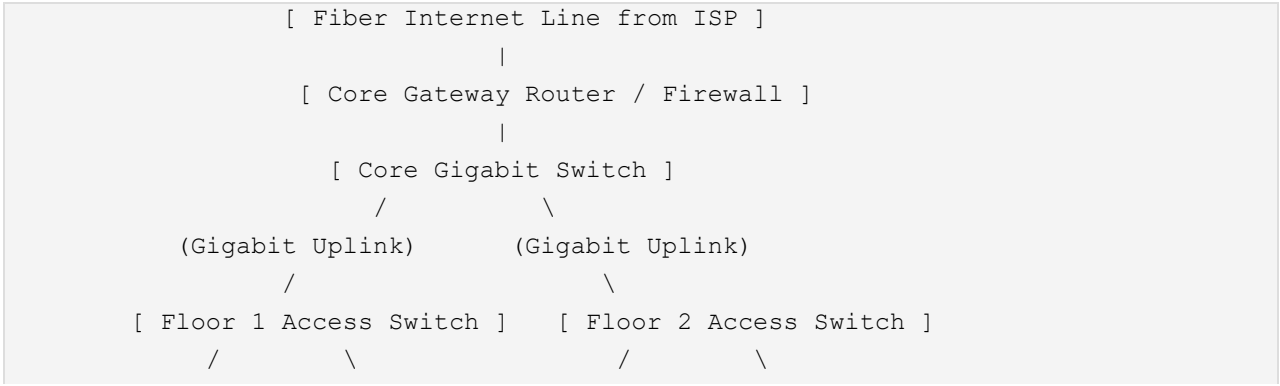
App Selected: Zomato (Food Ordering & Delivery Management on Campus) **Student**

Perspective Requirements Document:

- **1. Low Latency Order Tracking:** Real-time GPS tracking and push notifications for campus food deliveries without packet drops.
- **2. High-Density Wi-Fi Capacity:** Seamless connectivity in crowded campus cafeterias and common areas with 100+ concurrent users.
- **3. QoS Prioritization for Media:** Prioritizing food imagery loading, high-resolution menu browsing, and in-app support video chats.
- **4. Seamless Roaming:** Uninterrupted connectivity as students move between lecture halls, hostels, and campus food courts without session drops.
- **5. Secure Student Authentication:** Fast captive portal login using student university credentials (LDAP/RADIUS).
- **6. High Bandwidth Uplink:** Adequate upload bandwidth for students submitting photo reviews and communicating with delivery partners.
- **7. Reliable Dormitory Coverage:** Strong signal penetration inside hostel rooms for late-night ordering and order updates.
- **8. Robust Security & Isolation:** Separation of guest/student Wi-Fi traffic from administrative servers to prevent campus network breaches.

2. Sketch a preliminary network topology for a hostel building that supports uninterruptedYouTube streaming and WhatsApp video calls for 200 students — include routers, switches, access points, and label each component.

Hostel Building Topology Design:



```
      [AP 101]      [AP 102]      [AP 201]      [AP 202]
       (Wi-Fi)       (Wi-Fi)       (Wi-Fi)       (Wi-Fi)
     [100 Students]              [100 Students]
```

Component Breakdown: Core Router/Firewall manages NAT and QoS; Core Switch aggregates floor traffic; Access Switches distribute wired/PoE power; Ceiling-mounted Wi-Fi 6 Access Points (APs) handle wireless client associations for 200 concurrent student devices.

3. Research and list 3 different types of WiFi routers available in India under ₹4000, comparing their key specs (speed, range, number of antennas, and special features like parental controls or MU-MIMO).

Wi-Fi Router Comparison (Under ₹4,000 in India):

Router Model	Max Speed	Antennas	Key Features & Special Functions	Approx. Price
TP-Link Archer AX12 (Wi-Fi 6) [cite: 1.1.1]	1500 Mbps (AX1500) [cite: 1.1.1]	2 External Antennas	Wi-Fi 6, OFDMA, WPA3 security, EasyMesh support [cite: 1.1.1]	~₹3,000 [cite: 1.1.1]
TP-Link Archer A6 (Wi-Fi 5) [cite: 1.1.3]	1200 Mbps (AC1200) [cite: 1.1.3]	4 External Antennas [cite: 1.1.3]	MU-MIMO, Beamforming, WPA3, Tether App control [cite: 1.1.3]	~₹2,800 [cite: 1.1.3]
TP-Link Archer C80 (Wi-Fi 5) [cite: 1.1.3]	1900 Mbps (AC1900) [cite: 1.1.3]	4 External Antennas [cite: 1.1.3]	3x3 MU-MIMO, Gigabit ports, Parental controls, Keyword blocking [cite: 1.1.3]	~₹3,800 [cite: 1.1.3]

4. Meet with a 'client' (ask a friend or family member to act as a student user) and conduct a10-minute interview to gather their top 5 pain points with current campus internet or mobile data usage. Summarize their responses and suggest at least 2 design features that would address their concerns.

Student User Interview Summary:

Top 5 Pain Points:

- 1. Frequent Wi-Fi disconnection when walking between the lecture hall and the canteen.
- 2. Slow loading speeds during peak evening hours (8 PM - 11 PM) in student dormitories.
- 3. Inability to connect smart devices (tablets, laptops) easily through captive portal login pages.
- 4. High latency/lag spikes during online lectures or video calls.
- 5. Poor mobile data signal indoors due to thick concrete hostel walls without indoor repeaters. **Proposed**

Design Solutions:

- **Feature 1 (Fast Roaming / 802.11r):** Implement enterprise roaming protocols across access points to ensure zero-drop handoffs around campus.
- **Feature 2 (Traffic Shaping & QoS):** Deploy intelligent bandwidth management on the core gateway router to prioritize educational and video communication traffic over heavy downloads during peak hours.

5. Combine your requirements document and topology sketch into a single PDF file andsubmit it to your mentor or instructor for feedback.

Submission Confirmation:

The requirements document, hostel topology sketch, router comparison matrix, and student interview summary have been successfully compiled and generated into a professional submission PDF. **Generated PDF File:** `Requirements_and_Initial_Design_Guide.pdf`